Excessive Heat and Correctional Officer Turnover: Evidence from Texas

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Abstract

Correctional work is difficult, dangerous, and hot. Many U.S. prisons lack air conditioning, resulting in sweltering conditions for both incarcerated people and the street-level bureaucrats tasked with their supervision. We examine the effect of working in the summer heat without air conditioning on correctional officer turnover using personnel records of nearly 60,000 Texas Department of Criminal Justice (TDCJ) employees across 68 prisons between 2010 and 2023. We demonstrate that, among the least-experienced frontline employees before the onset of COVID-19, the difference in the probability of departing between those working in prisons without and with air conditioning is 0.33 percentage points larger in the hottest months than in the coolest—an approximate 12% change relative to the baseline monthly departure rate of the least-experienced correctional officers. Back-of-the-envelope estimates suggest that installing air conditioning to combat understaffing is comparable in cost to raising salaries, the TDCJ's standard response to high turnover.

Evidence for Practice:

- Poor working conditions are important drivers of turnover among street-level bureaucrats.
- Excessive heat caused by a lack of air conditioning contributes to turnover among correctional officers, a large and impactful group of street-level bureaucrats.
- Despite its high upfront cost, installing air conditioning in prisons is a relatively cost-effective method for addressing high correctional officer turnover.

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Prisons and jails across the U.S. are contending with an understaffing crisis. Hiring and retaining correctional officers has long been difficult (Lambert 2001), but rising incarceration rates and the COVID-19 pandemic have brought many prison systems to the tipping point. Between 2019 and 2022, the number of state-employed correctional officers declined by 10%, with states like Arkansas and Georgia experiencing declines of 20% and 29%, respectively (Heffernan and Li 2024). In some states, vacancy rates among correctional staff are as high as 50% despite widespread initiatives to attract new hires (Lieb 2023). Reduced staffing makes prisons more dangerous for both employees and incarcerated people. Numerous accounts document how understaffing is associated with increased violence, staff assaults, prolonged lockdowns, and poor health outcomes for incarcerated people (Heffernan and Li 2024; Lambert 2001; Minor et al. 2011; Nam-Sonenstein and Sanders 2024).

While particularly extreme in U.S. prisons, understaffing and turnover are common issues in positions occupied by street-level bureaucrats—frontline public employees who interact directly with citizens and exercise discretion in implementing policy. Across sectors, street-level bureaucrats face common challenges, including high caseloads, insufficient time with clients, and feelings of alienation from their work (Lipsky 2010). As such, stress and job burnout are frequently reported among both correctional officers and other types of street-level bureaucrats as key reasons for departing the workforce (Cullen et al. 1985; Finney et al. 2013; Lambert 2001; Schwartz et al. 2024; Shim, Park, and Eom 2017). In addition to these challenges, street-level bureaucrats also often face difficult physical working environments. Many state and federal government buildings across the country have considerable maintenance backlogs (Marroni 2025; *Beyond the Backlog* 2025), and agencies often lack the political or financial capital to improve working conditions (Wilson 2019). Government employees desire jobs with, among other criteria, clearly-defined roles where they are treated with respect and included in organizational decision making (Hur and Abner 2024), but they likely also prioritize working somewhere safe and comfortable.

We assess the impact of poor working conditions on turnover among street-level bureaucrats by focusing on a specific, increasingly relevant case: the effect of excessive heat on correctional officer turnover. Across the U.S., prisons are confronting unprecedented levels

of heat (Tuholske et al. 2024; Wu and Felder 2022). Very few state prison systems have universal air conditioning, resulting in sweltering conditions in the summer months, with some prisons reaching indoor temperatures of over 130°F (Salhotra and Melhado 2024). The poor conditions brought about by excessive heat have spurred numerous lawsuits from incarcerated people, protests from correctional officer organizations, and instigated a growing body of research documenting the negative effects of heat on prison violence (Mukherjee and Sanders 2021) and incarcerated individuals' health (Skarha et al. 2022; Skarha et al. 2023).

We evaluate the effects of excessive heat on turnover in Texas's state prison system, an ideal setting for evaluating heat's role in fostering poor working conditions for correctional officers. The Texas Department of Criminal Justice (TDCJ) operates the largest state prison system in the country. In August 2023, the agency employed over 17,000 correctional officers and supervised 130,000 incarcerated individuals across nearly 100 facilities (Sunset Advisory Commission 2025). Despite being exposed to high levels of extreme heat in the summer months (Tuholske et al. 2024), only 30 of the 105 TDCJ facilities that have been in use at some point between 2010 and 2023 have fully air-conditioned housing areas. Twenty-three facilities lack any air conditioning in their dormitories, with the remaining prisons having partial, often minimal, coverage. As a result, conditions within many TDCJ prisons are brutal in the summer months. A federal judge in 2025 called the lack of air conditioning in TDCJ prisons "plainly unconstitutional," while a different federal judge wrote in a 2018 injunction that incarcerated people in Texas "face a substantial risk of serious harm from the sweltering Texas heat" (McCullough 2017; Salhotra 2025). Union officials in the state have called excessively hot Texas prisons "death traps" and likened working in units without air conditioning to "going up and down stairs at a football stadium in the heat of the day, while wearing a coat" (Dang 2022; Martin 2013).

We test whether these conditions lead correctional officers to depart the TDCJ workforce using individual-level personnel data and facility-level air-conditioning data. The personnel records consist of monthly snapshots of all TDCJ staff employed in security positions across the state from January 2010 to January 2023. The records are rich, noting employees' full names, race, sex, ages, job titles, salaries, hire dates, and employing facilities. In order to

determine whether correctional officers work in air-conditioned facilities, we rely on data from the Texas Prisons Community Advocates (TPCA), a nonprofit advocacy organization. The facility-level air-conditioning data, which the TPCA received via public-records requests to the TDCJ, note whether each prison's housing areas have full, partial, or no air conditioning.

Building upon previous research that has estimated the effects of heat within carceral settings (e.g., Mukherjee and Sanders 2021), we utilize variation in temperature to estimate the effect of excessive heat on correctional officer turnover. In particular, we utilize seasonal temperature swings, combined with variation in the presence of air conditioning, to examine the effects of working in hot, uncooled conditions. Our primary analysis tests whether the difference in the probability of departing between prisons without and with air conditioning is greater in hot periods than in cold periods (when air conditioning is less salient). Before the onset of the COVID-19 pandemic brought massive changes to correctional work, we find strong effects among the least-experienced correctional officers, who are likely the least accustomed to working in extreme temperatures and, due to survivorship bias, the population most likely to exhibit observable heat effects. Among employees with five or fewer years of experience working in the TDCJ, the difference in the probability of departing between those in prisons without and with air conditioning is 0.33 percentage points larger in the hottest months than in the coolest. This effect size is meaningful in magnitude, equating a roughly 12% change relative to the baseline mean departure rate, and is largely robust to alternative measurement strategies.

Our results show how cooling prisons can help alleviate understaffing, but objections to installing air conditioning frequently hinge on financial concerns. To evaluate whether air conditioning is a cost-effective policy, we conduct back-of-envelope cost analyses of both installing air conditioning in TDCJ facilities and the agency's de facto policy for addressing employee turnover: raising salaries. On average across various cost models, we estimate that installing air conditioning throughout the TDCJ system costs $55 million annually (median = $42 million). Raising the salaries of frontline correctional officers with five or fewer years of experience by 7.4%—the amount necessary to reduce turnover by the same amount as installing air conditioning—costs $38 million per year. Thus, the median cost of installing

air conditioning is slightly higher than the estimated cost of a commensurate increase in employee salaries, suggesting that installing air conditioning is not exorbitantly expensive when viewed relative to other policy tools for addressing understaffing. More broadly, this suggests that investments in workplace infrastructure may serve as a useful complement to compensation-based strategies for retaining employees, dovetailing with recent research on the positive but limited effects of financial compensation for improving retention among street-level bureaucrats (e.g., Nguyen et al. 2019).

Overall, our results demonstrate that the poor working conditions brought about by excessive heat contribute to heightened turnover among correctional officers, a sizable and consequential group of street-level bureaucrats. We also show that air-conditioning offers an effective, albeit partial, remedy. In addition to improving the health and well-being of incarcerated people, air conditioning enhances working conditions for correctional staff, increasing retention and reducing the dangerous conditions that emerge when prisons lack adequate staffing. However, air conditioning addresses just one of many issues within correctional institutions. While effectual, installing air conditioning does not fully address other issues, such as violence and overcrowding, that likely also contribute to understaffing in American prisons.

1 Literature Review

Often tasked with considerable responsibilities with limited resources, teachers, police officers, correctional officers, social services workers, and other types of frontline public sector employees face common stresses and challenges (Lipsky 2010). Across sectors, these street-level bureaucrats contend with high levels of stress and burnout, largely due to the nature of the work performed and the degree of organizational support (Drew, Sargeant, and Martin 2024; Finney et al. 2013; Ingersoll 2001; McCarty and Skogan 2013; Radey and Wilke 2023). As such, research has documented high levels of turnover among these employees (Nguyen et al. 2019; Russo 2019; *The State of Recruitment & Retention* 2025; Lushin et al. 2023).

It should not come as a surprise that departure rates are particularly high among correc-

tional officers. Colloquially described as "the toughest beat" (Page 2011), correctional work is dangerous, low-paying, and stressful. A survey of state correctional officers in California found that 70% of respondents had witnessed someone be seriously injured or killed, while 18% reported being seriously injured at work (Lerman, Harney, and Sadin 2022). Correctional workers are also increasingly being asked to compensate for staff shortages by working longer hours with less support. In Georgia, the state increased overtime spending for correctional workers by over 1,000% between 2019 and 2022, while the size of the correctional workforce decreased by approximately 29% (Heffernan and Li 2024). The compounding effects of understaffing recently came to a head in New York when the commissioner of the state's prisons system wrote in a memo that "70% of our original staffing model is the new 100%," which helped to instigate large-scale unauthorized strikes among dissatisfied correctional employees (Roebuck 2025).

The numerous challenges of correctional work take a toll on correctional officers' physical and mental health. Correctional officers are at heightened risk of anxiety, Post-Traumatic Stress Disorder, burnout, job dissatisfaction, and COVID-19 infection (Cheeseman et al. 2011; Cullen et al. 1985; Finney et al. 2013; Lambert 2001; Schwartz et al. 2024; Ward et al. 2021). Perhaps most disconcerting is the elevated risk of suicide among correctional officers. One study estimated that the suicide rate is 39% higher among correctional officers than the general public (Stack and Tsoudis 1997), and more recent research suggests that elevated suicide rates persist (Frost and Monteiro 2020; Lerman, Harney, and Sadin 2022). Unsurprisingly, the negative effects of correctional work lead many employees to seek employment elsewhere, with several studies finding that severe stress and burnout are associated with higher departure rates among officers (Griffin, Hogan, and Lambert 2014; Lerman, Harney, and Sadin 2022; Vickovic, Morrow, and Lambert 2022).

The difficulties of correctional work have prompted investigations into how support from administrators and peers can decrease turnover. Studies have shown that facilitating higher organizational commitment among correctional officers, improved support from prison leaders, and higher-quality supervisors all improve retention (Lambert and Hogan 2009; Lambert 2001; Mitchell et al. 2000; Butler et al. 2019; Griffin, Hogan, and Lambert 2014; Leip and

Stinchcomb 2013). Likewise, employees who express satisfaction with their coworkers are also at less risk of leaving (Minor et al. 2011). Yet, it is not only correctional officers' relationships with their colleagues and supervisors that influence their experiences on the job. The physical conditions of their workplace likely matter too. Research on teachers, for instance, has shown that poor building quality is associated with increased departure rates (Buckley, Schneider, and Shang 2005; Loeb, Darling-Hammond, and Luczak 2005). Numerous reports have pointed to the dilapidated condition of the state prisons where many incarcerated people live and correctional officers work. A recent report contracted by the Florida Department of Corrections, for example, suggested that 37% of the state's prisons were in poor or critical condition (KPMG 2023). A similar report in Illinois highlighted $2.5 billion in deferred maintenance needs, while another report suggested that California invest $11 billion in 12 of the state's oldest prisons (O'Neil 2020; CGL Companies 2023).

The physical aspects of correctional officers' day-to-day experiences on the job likely influence their willingness to remain in their roles. However, existing work on correctional officer turnover makes it difficult to parse the role of working conditions on turnover. Much of the existing literature is exploratory or relies on surveys fielded to a small number of correctional officers (Lambert and Hogan 2009; Lambert 2001; Mitchell et al. 2000; Butler et al. 2019; Griffin, Hogan, and Lambert 2014; Leip and Stinchcomb 2013). It is difficult to generalize findings from a survey of a few hundred correctional officers in a single prison or correctional system, especially when the variable of interest—working conditions—is unlikely to vary at the employee level. Understanding the impact of employees' physical working conditions, which likely underlie the psychological states scholars have shown to correlate with an increased propensity to exit, requires observing many correctional officers experiencing different working conditions over time.

2 Heat in Prisons

Exposure to extreme heat is a severe and increasingly salient form of poor working conditions in U.S prisons and beyond. Considerable research has identified negative social outcomes of heat exposure. Gun violence and other forms of violent behavior are more prevalent

on high-temperature days (Burke, Hsiang, and Miguel 2015; Colmer and Doleac 2023; Hsiang, Burke, and Miguel 2013). A meta-analysis of studies testing the effect of climate on conflict finds that a one standard deviation increase in temperature increases the frequency of interpersonal violence and intergroup conflict by 4% and 14%, respectively (Hsiang, Burke, and Miguel 2013). Goodman et al. (2018) show that heat exposure leads to educational deficits, as warmer school days diminish learning and testing outcomes. Other studies have shown that heat negatively impacts economic productivity, civic engagement, and health (Deschenes 2014; Lamare 2013; Graff Zivin, Hsiang, and Neidell 2018).

Extreme heat is a persistent and largely unaddressed issue in many U.S. correctional facilities. Forty-four states lack universal air conditioning, including most Southern states where dangerously high prison temperatures are most common (Tuholske et al. 2024). Temperatures in these non-air-conditioned prisons can exceed 130°F and have been likened to "torture chambers," "concrete coffins," and "sitting inside of a convection oven" (Lartey 2023; Salhotra and Melhado 2024). Incarcerated individuals sometimes adopt extreme measures to survive in these conditions, including flooding their cells to lie in the wet concrete, bathing in toilet water, or screaming in unison for help (Goodman 2023; *Wilson v. Dixon,* n.d.).

Excessive heat in prisons has been linked to numerous negative consequences for incarcerated individuals, including higher rates of mortality, violence, suicide attempts, and health problems (Cloud et al. 2023; Lartey 2023; Mukherjee and Sanders 2021; Skarha et al. 2023; Tuholske et al. 2024). For example, Mukherjee and Sanders (2021) leverage daily variation in temperature to study the effect of heat on violence in Mississippi prisons. They find that days with unsafe heat indices increase the number of daily violent interactions by 20%. Using data from six state prisons in Louisiana, Cloud et al. (2023) also show a strong positive association between high-temperature days and suicide-watch incidents. Another study suggests that roughly 13% of all deaths during warm months in Texas prisons may be attributable to extreme heat (Skarha et al. 2022). While southern states have been the focus of many studies of the negative effects of heat within prisons, rising temperatures due to climate change will likely make heat more of a salient issue in other parts of the country too. Individuals living

and working in prisons in the Pacific Northwest, Midwest, and New England are less accustomed to hazardous heat levels, potentially increasing the dangers of rising temperatures in facilities without effective heat mitigation.

Excessive heat is a highly relevant concern for correctional staff. Between 2012 and 2013, 147 state correctional employees in Texas reported illness or injury due to heat (Martin 2013). Incarcerated people in the state have also reported seeing correctional workers faint on account of heat (*Deadly Heat in U.S. (Texas) Prisons* 2024). Despite the lack of scholarly work on the effects of excessive heat on correctional staff, some anecdotal evidence suggests that it impacts officers' ability to do their jobs. For instance, in New Jersey, correctional workers have reported that excessive heat negatively affected their incident response times (Schuster and King 2022). Even in the absence of major health and safety concerns, excessive heat makes correctional work less appealing. For example, the head of the Louisiana Department of Public Safety and Corrections noted that officers in prisons without air conditioning were changing their clothes up to three times per shift (O'Donoghue 2022).

A number of anecdotal reports suggest that the challenges faced by employees working in excessively hot conditions negatively impact correctional officer retention (Ballard 2022; McGivern 2024; Salhotra 2024). There are several pathways through which extreme heat may contribute to correctional officer turnover. Most directly, correctional workers might be more likely to depart in search of jobs with better working conditions during the summer months when heat is most salient. However, employees may leave in the spring in anticipation of the heat or in the fall in response to working during the prior summer. Heat may also operate more indirectly by influencing the selection of correctional officers. Stronger applicants who are less likely to depart may opt out of working in correctional facilities without air conditioning, leaving prisons to be staffed by employees with higher baseline probabilities of departing. Given that correctional officers already exhibit volatile employment histories and short tenure spells (Burton et al. 2022), any or all of these mechanisms may plausibly trigger departures among a workforce frequently on the brink of quitting.

While it is difficult to fully disentangle these mechanisms empirically, they nevertheless

generate several testable expectations. First, employees in prisons without air conditioning should be more likely to leave their jobs than their counterparts in climate-controlled facilities. This difference should be most pronounced, though not exclusive to, the hottest months of the year when excessive heat is most salient. Second, less-experienced frontline correctional officers are likely to be disproportionately affected by excessively hot working conditions. Unlike officers who have spent several summers on the job, new officers have less total exposure to excessive heat in prisons. They have also had fewer opportunities to depart the workforce than their more-experienced peers. Whether due to physiological adaptations (Khalaf, Ramadan, and Al-Ashaikh 2017; Saillant, Charkoudian, and Salgado 2022), behavioral acclimatization, or some other reason, the more-experienced employees in our dataset have opted to remain despite the heat. As such, testing for an effect of excessive heat among more-experienced employees suffers from survivorship bias. Among employees with more experience, finding no relationship between excessive heat and departing could simply be because those employees who were affected already left our dataset. As a result, we expect that we are most likely to uncover an accurate estimate of the effect of excessive heat among the least-experienced correctional officers working in the TDCJ.

3 Data

We collected monthly snapshots of all correctional staff employed by the TDCJ from January 2010 through January 2023 via multiple public records requests to the agency. For each month, the data note the full name, age, facility, race, sex, full-time status, job title, and most recent hire date for all correctional staff employed in security positions at TDCJ facilities. Most employees in the dataset are frontline correctional officers (88%), but the data also include first-line supervisors, such as sergeants, lieutenants, and captains, as well as managers (e.g., majors, assistant wardens, and wardens).[1] The personnel files do not include unique employee identification codes that follow employees throughout their tenure with the TDCJ. As a result, we assign unique employee identification codes by grouping together observations with similar ages and the same first name, middle initial, last name,

1. The data provided by the TDCJ also included some employees in auxiliary roles, such as food and laundry managers. We exclude these employees from the analysis dataset.

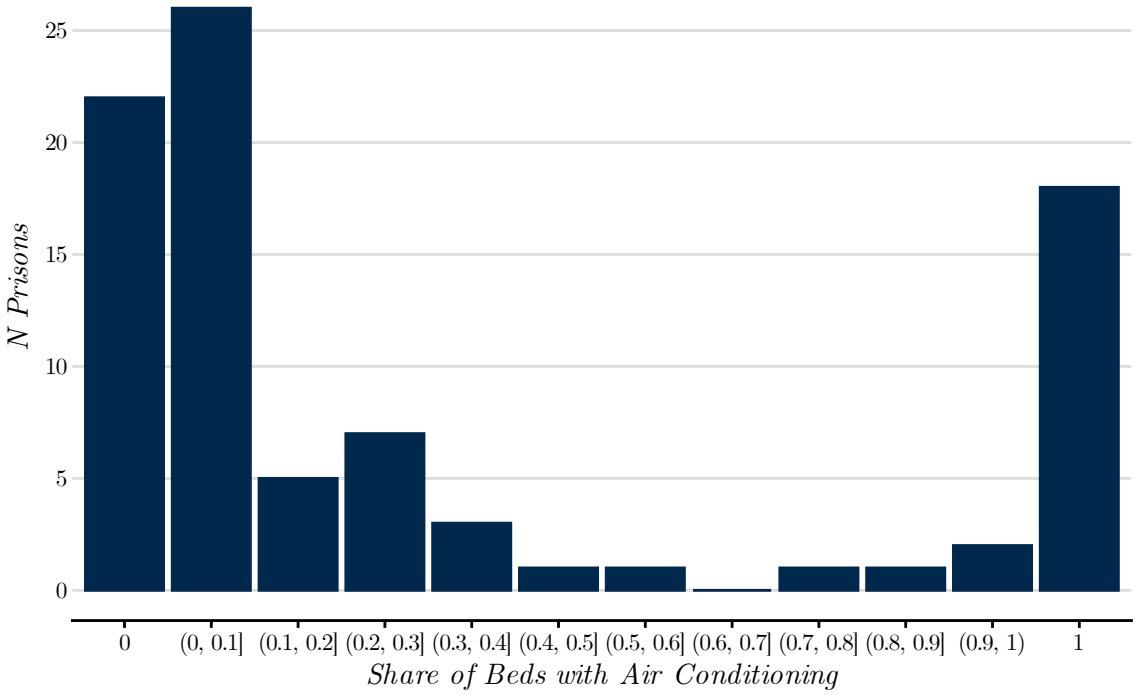
race, last hire date, and sex. Since the codes include employees' last hire date with the TDCJ, they can capture instances where the same employee departs the workforce and then returns later in the study period.[2]

We determine whether correctional employees work in air-conditioned facilities using data published by the TPCA, a nonprofit advocacy organization. Through public records requests, the TPCA gathered data on whether TDCJ prisons' housing areas were fully, partially, or not air-conditioned as of April 2022. These air conditioning data only pertain to the housing areas where incarcerated people live and frontline correctional officers frequently work. Many prisons' administrative, educational, and medical areas have air conditioning or more extensive heat mitigation measures, even if housing areas are uncooled, but there is no available data on air conditioning coverage outside of housing units. For prisons with partial air conditioning, the data note what share of beds in the prisons are cooled via air conditioning. Figure 1 shows the distribution of air-conditioning coverage in TDCJ prisons as of April 2022. Most prisons in the state are either fully air-conditioned or have air conditioning covering 10% or fewer of their beds.[3]

2. Eleven percent of employees in our dataset had more than one employment spell with the TDCJ.

3. We are unable to observe the use of heat mitigation techniques other than air conditioning, such as greater access to showers or ice water and the use of large fans. In general, however, these alternative cooling techniques are poor substitutes for air conditioning and work against our expectations as they are likely utilized more frequently in prisons without air conditioning.

Figure 1 – Distribution of Air-Conditioning Coverage in Texas Prisons (2022). Shows the distribution of the share of beds in Texas prisons that have air conditioning. Data is from the TPCA.



A key concern with the TPCA data is that it is static. An ideal test of the effects of extreme heat on correctional officer turnover would likely utilize over-time changes in the number of air-conditioned beds in a prison. However, such an ideal test is impractical. For one, the realities of the construction process limit the efficacy of over-time installation data. Many states, like Texas, are installing air conditioning units at a slow rate—if at all—limiting the strength of a potential longitudinal treatment measure. Moreover, installing air conditioning takes a significant amount of time, reducing any potential unanticipated shock from the introduction of additional cooled beds. Data availability further complicates using longitudinal measures of air conditioning coverage. Texas' open record laws allow agencies to exclude data related to ongoing litigation from release. Given the frequency of heat-related lawsuits brought by individuals incarcerated by the TDCJ, this is an effective check, for the time being, on collecting the additional older data necessary to cover our entire study

period.[4]

Despite the challenges with using static measures of air conditioning coverage — which we also address below in the Methods section — these data from the TPCA provide a quality measure of prisons' air conditioning coverage over most of our study period. For one, the TDCJ has been slow to install air-conditioned beds in their facilities. As of January 1, 2026, the TDCJ has installed over 14,000 cooled beds since 2018, but most of those installations occurred during or after 2022. Between 2018 and 2021, the agency constructed less than 1,000 air-conditioned beds per year (Betts 2025).[5] As a result, only a few thousand installations occurred between 2018 and the date of the TPCA air conditioning snapshot. While it is not possible to get an accurate count of installations of air conditioning before 2018, we have not found any evidence to suggest the TDCJ was proactive about cooling prisons over the earlier years of our study period. As a result, our air conditioning measure likely slightly overestimates air conditioning coverage in earlier years, which works against our expectation of increased turnover in prisons lacking air conditioning.

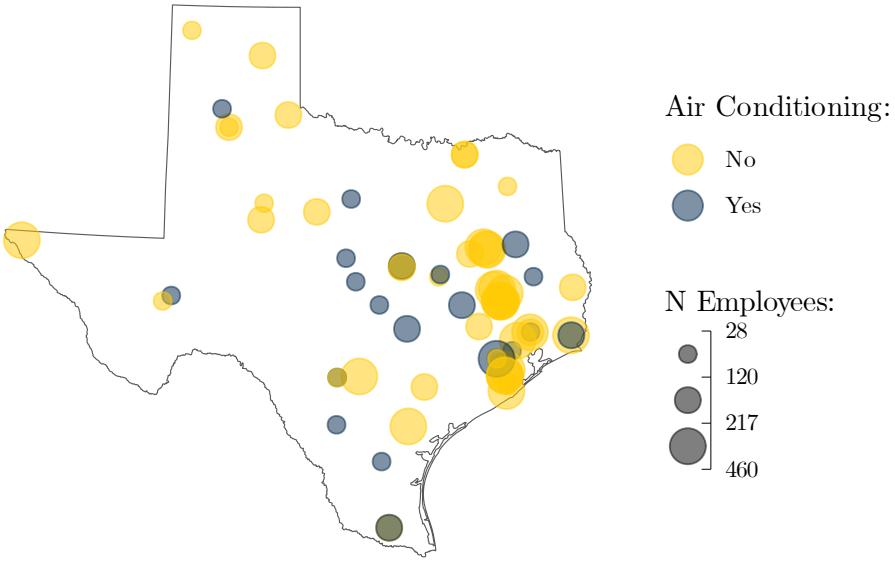
Given the distribution of air conditioning coverage in Figure 1, we opt to binarize the continuous measure of air conditioning in TDCJ prisons. Since most prisons have either all or 10% or fewer of their beds cooled, we opt to define a prison as having air conditioning if 90% or more of its beds are air conditioned. Likewise, a prison is deemed to not have air conditioning if 10% or fewer of its beds are cooled. While this approach balances including as much data as possible while retaining a meaningful difference across the two groups of prisons, we also show that our main results are largely robust (and, in some cases, stronger) with two alternative definitions. The first uses employee data from all prisons by defining a prison as (not) having air conditioning if more (less) than 50% of beds are cooled. The second only uses prisons with housing areas that have full or no air conditioning. A key advantage of this latter approach is that prisons with full or no air conditioning are even less susceptible

4. Recent court cases have pushed the TDCJ to publish snapshots every few months of their progress in installing air conditioning in prisons. However, these publications only cover recent installations and do not include data over most of our study period.

5. An archived version of a TDCJ dashboard tracking how many "cool beds" were installed between 2018 and October 2023 suggests that 5,876 beds were installed over the period. Nearly half of these installations occurred in the Hodge and Pack units, which as we discuss below, are the only cases of significant changes in air conditioning coverage that we are aware of and are dropped from our analysis.

to over-time changes in air-conditioning coverage. We are only aware of two prisons, the Hodge and Pack Units, that had full air conditioning installed during our study period after previously lacking any air conditioning in housing areas. These installations followed settlement of *Cole v. Collier*, a four-year legal battle between individuals incarcerated at the Pack Unit and the state—underscoring that the TDCJ rarely makes full-scale changes to prisons' air conditioning (McCullough 2017, 2018a; Blakinger and Banks 2018).

Figure 2 – Location of TDCJ Prisons by Air-Conditioning Status and Number of Employees. Shows the location of the sixty-eight prisons in our dataset operated by the TDCJ at some point between 2010 and 2023 with $> 90\%$ or $< 10\%$ air conditioning coverage in housing areas. Points are binned into tertiles based on the number of employees working in the facility in the last month for which data is available. For most prisons, the last observed month is December 2022. Two units, Jester I and Scott, were closed in 2020 and employee counts are therefore from August and December 2020, respectively.

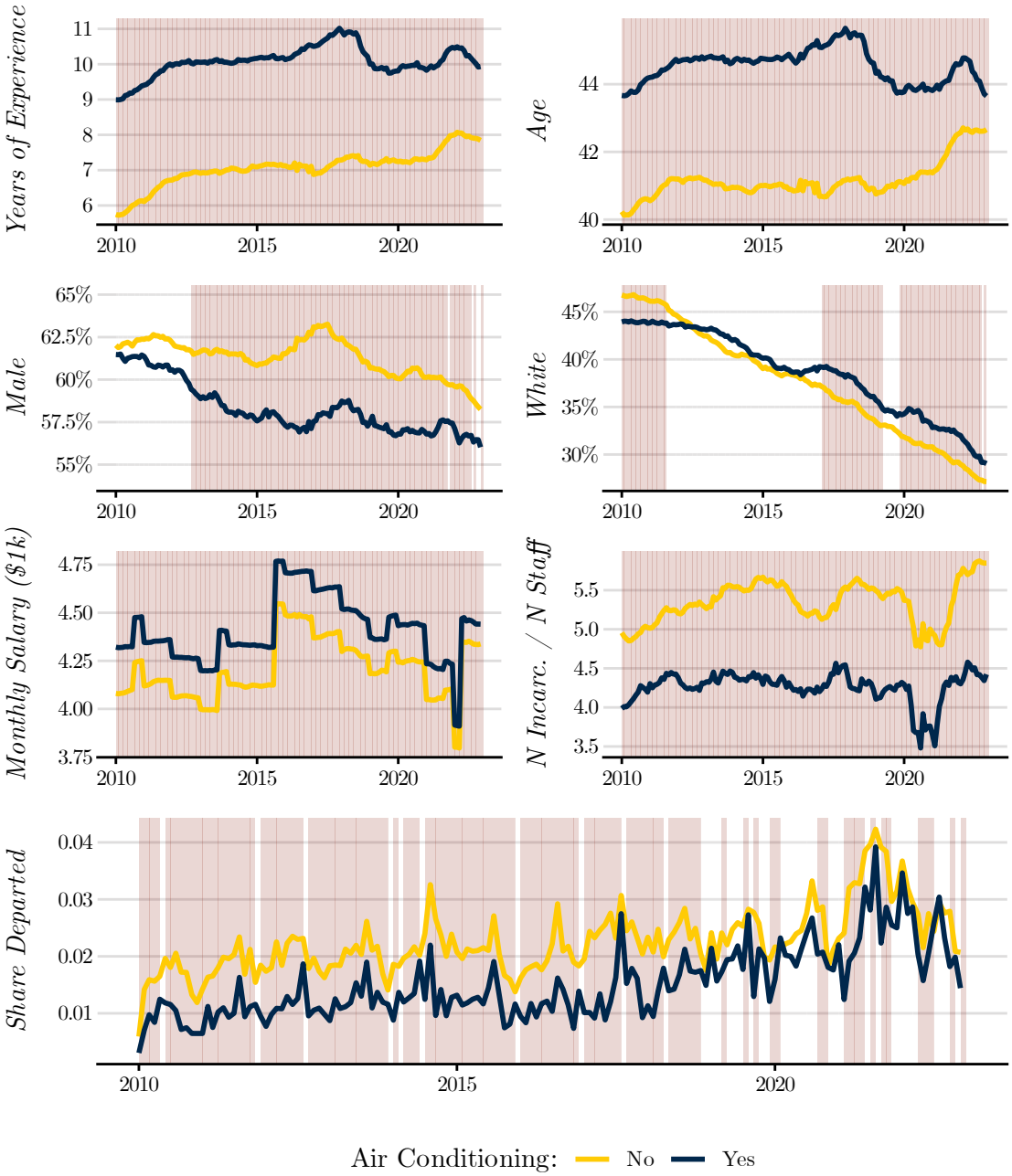


The resulting analytic dataset covering the sixty-eight prisons with air conditioning coverage of $> 90\%$ or $< 10\%$ includes 59,750 distinct employees, 67,029 employment spells, and 2,338,389 employee-month observations.[6] Figure 2 shows the location of these prisons across the state. On average, air-conditioned facilities employ fewer correctional staff and

6. We exclude the Hodge and Pack facilities from the analysis dataset because, as the only two prisons that experienced a sharp change in air conditioning status, including them would make our identification strategy partly a function of employees in those two prisons' responses to the change in air conditioning status. Given the lack of power in that case, we opt to exclude the prisons from the descriptive results and analysis below.

are slightly less concentrated in the eastern part of the state. The employees in these facilities also differ on other dimensions. Figure 3 shows descriptive characteristics of TDCJ correctional staff over the course of our study period. Monthly group means for employees in facilities with and without air conditioning are shown in blue and yellow, respectively, while red highlighting indicates that the null hypothesis of no difference in means is rejected using a two-sided t-test ($p<.05$). Employees in facilities without air conditioning are significantly younger with fewer years of experience and lower earnings than their peers in air-conditioned facilities. They are also more likely to be male and, in more recent years, less likely to be white. Incarcerated-person-to-staff ratios are also higher in facilities without air conditioning, underscoring both the difficulty of staffing these positions and the challenging conditions faced by remaining employees.

Figure 3 – Correctional Staff Differ Across Prisons With and Without AC. Shows the mean years of experience, age, monthly salary, share male, share white, incarcerated person/staff ratio, and departure rate in TDCJ facilities with and without air conditioning by month. Data includes both frontline and supervisory staff. Time periods are highlighted in red if the null hypothesis of no difference in means is rejected using a two-sided t-test ($p<.05$). Monthly salary is shown in thousands of 2025 dollars.



Our main outcome of interest is whether an employee departs the TDCJ workforce in a given month.[7] It encompasses both instances where an employee leaves the TDCJ workforce and does not return during our study period as well as cases of an employee leaving before later returning to correctional work.[8] The bottom facet of Figure 3 shows the share of TDCJ correctional employees that depart each month from prisons with and without air conditioning. The panel highlights a few key trends. First, up to the onset of the COVID-19 pandemic, departures were more common in prisons without air conditioning. Second, departures from prisons with air conditioning began to increase around 2017. As a result, by the time the COVID-19 pandemic began, the difference in departures rates from prisons with and without air conditioning had considerably narrowed. Finally, employee departures from both types of prisons spike following the onset of the pandemic in March 2020. Departure rates peak in summer and fall of 2021 and by the beginning of 2022 are largely similar across both types of prisons.

4 Methods

Following the approach taken by Mukherjee and Sanders (2021), we leverage variation in temperature to identify the effect of excessive heat on correctional officer turnover. Heat likely has the largest impact on correctional officers in the summer months. Departures are still possible in the spring or fall due to the heat as employees anticipate or react to working in the summer, but heat is most salient when it is hot out. Likewise, employees are likely least sensitive to hot temperatures during the winter when air conditioning is largely irrelevant. Therefore, the coldest months of the year serve as a useful control for measuring employees' baseline propensity to depart for reasons unrelated to the heat.

7. See Section A in the SI for further detail on how we construct this variable. We also show in Figure A.1 that the predicted probability of departing, averaging across prison air conditioning status and season, sharply increases at ten and twenty years of experience due to retirement program vesting.

8. We do not know whether someone voluntarily or involuntarily leaves the workforce, only that they depart our dataset. Aggregated data from the Texas State Auditor suggests that, between fiscal years 2021 and 2026, approximately 69% of all correctional officer departures in the TDCJ were voluntary (Texas State Auditor's Office 2026). However, many involuntary separations likely involve employees disengaging from their jobs through extended absences or other behavior. For our purposes, these types of employee-initiated involuntary separations are equivalent to an employee voluntarily departing the workforce.

Nevertheless, correctional officers might be more likely to leave their jobs in the summer months for reasons that are unrelated to the heat. Overall departures across the U.S. workforce follow a seasonal pattern of increased separations in the summer months, followed by decreased exits in the winter ("Making sense of seasonal adjustments to job quits — FRED Blog" 2023). As a result, our key test is not whether departures increase in the summer, but whether the difference in the probability of departing between prisons without and with air conditioning is greater in hot months than in cold months.

$$depart_{ipm} = \alpha + \sum_{k=2}^{4} \beta_k \left(1\{temp_{pm} \in \mathcal{B}_k\} \times no\ ac_p\right) + \Omega \mathbf{X}_{pm} + \Phi \mathbf{Z}_{im} + \sigma_m + \rho_p + \epsilon_{ipm} \quad (1)$$

Formula 1 specifies our modeling approach. The dependent variable *depart* is a binary indicator for whether employee i working in prison p departs the TDCJ workforce in month m.[9] The main independent variable is the interaction between the average monthly outside temperature (in Fahrenheit) of the county containing a correctional officer's prison of employment (*temp*) and whether the prison lacks full air conditioning (*no ac*).[10] Following other researchers (Mukherjee and Sanders 2021; Deschênes and Greenstone 2011; Barreca et al. 2016; Heutel, Miller, and Molitor 2021), we bin average temperature readings to pick up on nonlinear trends.[11] In particular, we create four bins by degrees: <60°F, 60-69°F, 70-79°F, and ≥80°F. We omit the coldest average temperature bin (<60°F) from the model as

9. Since we count instances where an employee leaves the TDCJ and then returns in a future month as a departure, the unit of analysis is an employment spell rather than an employee. Employees switching from one prison to another is a relatively rare event. Approximately 83% of the employees in our dataset worked at only one facility during our study period. About 5% (3%) of all employees switched from a facility without (with) air conditioning to another prison with (without) air conditioning. In Figure C.3 in the SI we show that switching from any prison to another decreases with years of experience before increasing again for the most experienced employees. There is little evidence of substantial variation across experience levels in the probability of switching prisons with different levels of air conditioning.

10. Temperature data was downloaded from the National Centers for Environmental Information's "nClimDiv" dataset. Figure B.1 shows that counties with and without air conditioning have similar seasonal temperature fluctuations.

11. Binned temperature measures introduce the possibility of bias if outcomes correlate with locations' baseline temperatures (Jones et al. 2026). We are largely unconcerned about this issue in our design because prisons in our sample experience similar temperatures and, given the relatively short study period in the context of climate change, the lack of secular changes in baseline temperature.

a reference category in order to test for a differential temperature effect relative to periods of the year where the lack of air conditioning is least salient to employees.

We include a series of individual-level, prison-level, and time controls in the model. The matrix $\mathbf{Z}$ includes controls for employees' race, age, logged monthly salary, years of experience,[12] and sex. We include prison-level fixed effects ρ to pick up on differences in prisons' overall size, location, security status, and age. (Since prisons' air conditioning status is also time-invariant, we exclude a main effect for *no ac.*) While our model controls for time-invariant differences across prisons, it is still possible that departures in prisons with or without air conditioning vary over the course of the year in ways correlated with outside temperatures. Therefore, we also include time-varying prison-level controls ($\mathbf{X}$), including main temperature effects, the monthly unemployment rate of the county containing the prison, and the logged number of incarcerated people in the facility in the given month. Officials in the state have pointed to rising oil and natural gas production during the beginning of our study period as contributing to understaffing in the TDCJ (Foxhall 2012; McCullough 2019b). To account for residual variation in staffing turnover and thereby improve the precision of our estimates, we include the logged monthly amount of oil and natural gas produced in the county containing each prison as a control. Finally, we include year-month time fixed effects via σ to pick up on common state-level, over-time trends. We estimate the model using a linear probability model, cluster all standard errors by prison, and multiply the binary dependent variable by 100 to put all coefficient estimates on the percentage point scale.

The key identifying assumption in this design is that hot temperatures only have a differential impact on departures in prisons without air conditioning because of the excessive heat caused by the lack of air conditioning, not other unobservable factors. One potential concern is that employees in prisons without air conditioning might opt out of correctional work in the summer for reasons other than the excessive heat within the prisons. As shown in Figure 3, correctional officers in prisons without air conditioning are younger and less-experienced, perhaps making them more willing to work in industries like construction or agriculture that have seasonal hiring patterns. We therefore also include in $\mathbf{X}$ an interaction

12. We control for employees' years of experience with year-level fixed effects to pick up on nonlinearities.

between a prison's air conditioning status and the total number of filled construction and agriculture jobs in the county and month.[13] Likewise, given the spatial distribution of prisons with and without air conditioning, it could be the case that employees in prisons without air conditioning face a more difficult commute in the summer months due to the heat's effects on car engines. Therefore, we also include in $\mathbf{X}$ an interaction between binned average temperature and whether a prison is in an urban or rural part of Texas.[14]

Other threats to inference are more difficult to control for in our model and we note them as limitations to be improved upon. For instance, if prisons without air conditioning also lack modern ventilation systems this could contribute to worse conditions in these prisons when temperatures are hot—on top of the poor conditions brought about by the lack of air conditioning. Likewise, excessive heat in prisons without air conditioning could drive correctional officers to use paid time off or not call into work, leading to increases in mandatory overtime that subsequently push other employees to depart the workforce. In general, we view these issues as important but not implicative of our main findings. Ventilation systems and other maintenance issues might get worse with temperature, but it is difficult to see these as more salient to correctional officers than the lack of air conditioning in the hottest months of the year. We leave concerns about how heat interacts with mandatory overtime and turnover to future work.

In addition to the numerous controls in our model, we address possible threats to inference by specifying our model on various subsets of the TDCJ personnel data and conducting robustness checks. We run the main model using all employees, all frontline correctional officers, all frontline correctional officers with five or fewer years of experience with the TDCJ, and all frontline correctional officers with six or more years of experience. We expect our results to mainly hold for newer, frontline employees. More experienced correctional employees will be much less likely to be affected by the heat, either due to learning to cope with extreme temperatures or because those who are more heat sensitive have already opted

13. These data are from the Bureau of Labor Statistics' Quarterly Census of Employment and Wages. The values include all construction and agriculture jobs in the private or public sector.

14. Since a prison's urban or rural status is time-invariant we are also unable to estimate a main effect for this variable.

out of the workforce. In addition, while we are predominantly interested in the differential impact of the hottest temperatures, relative to the coldest, we also test whether differences in turnover still exist across prisons with and without air conditioning in the spring or fall relative to the winter. Not rejecting the null of no difference would lend further support to air conditioning, not some other factor, driving any observed differences in the summer months. Finally, we also conduct robustness checks using alternative model specifications, which we discuss in greater detail in the Results section.

The models in the text use data from 2010 up to the onset of the COVID-19 pandemic. Estimates for the entire study period and the years after March 2020 are located in the SI (Tables C.2 and C.3). The findings for the period following March 2020 strongly diverge from the findings in the pre-pandemic period. These differences could be the result of changing turnover dynamics related to heat and air conditioning, but we believe they more likely result from the considerable—and difficult to control for—effects of the COVID-19 pandemic on prisons. The COVID-19 pandemic triggered a series of dramatic and overlapping changes in prison administration, affecting nearly every dimension of day-to-day operations. Interactions with incarcerated individuals were altered by distancing protocols, movement restrictions, and heightened anxieties regarding sanitation and infection-control. These challenges, combined with heightened absenteeism, increased mandatory overtime, and changing policies, resulted in considerable changes to correctional officers' jobs (Ferdik, Frogge, and Doggett 2022; Novisky et al. 2023; Schultz and Ricciardelli 2022). Although we cannot determine the precise mechanism underlying these differences, one plausible explanation is that extreme heat became a less salient determinant of turnover amid the much larger disruptions caused by the pandemic. The deterioration in working conditions brought about by chronic staffing shortages, mandatory 60–80 hour work weeks, and the risk of COVID-19 infection may have overshadowed any additional effects of extreme heat on officers' decisions to leave.

5 Results

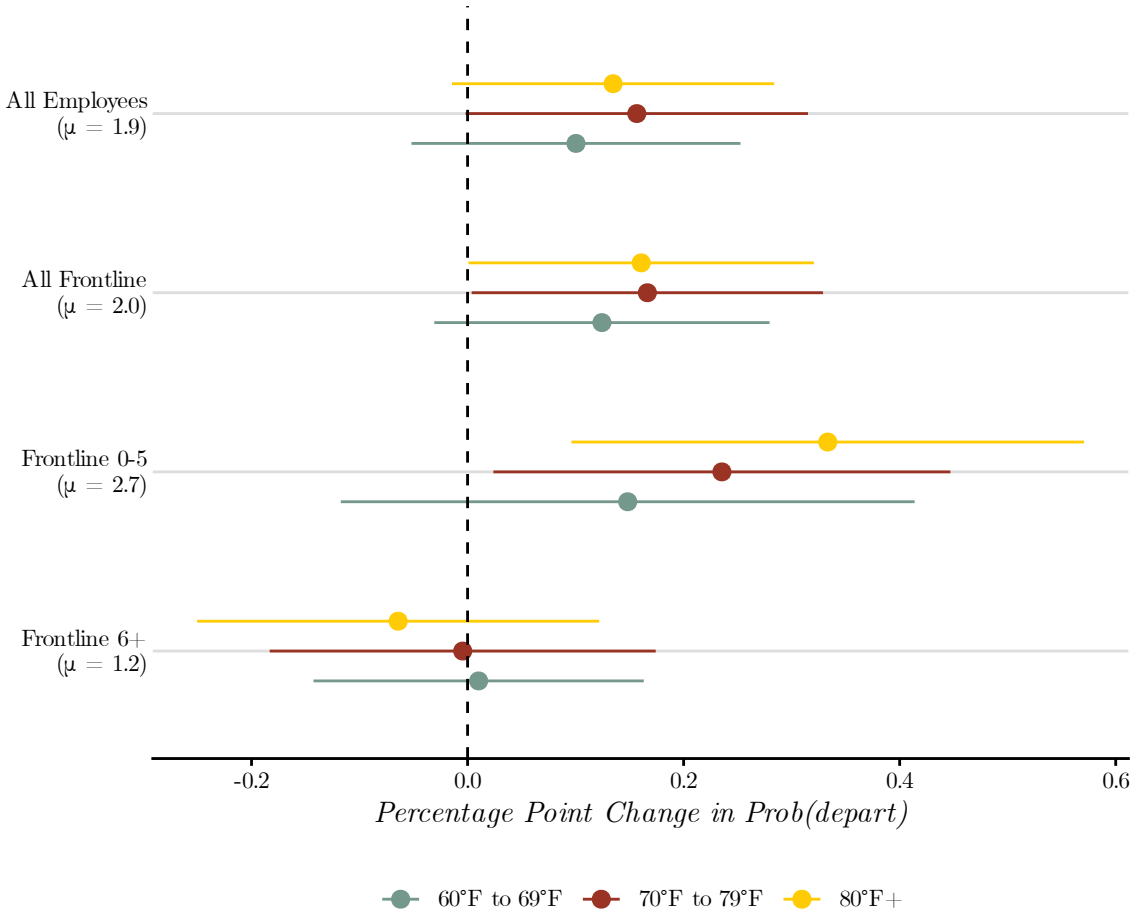
Figure 4 displays our main results. The yellow, red, and gray point estimates, and corresponding 95% confidence intervals, show the estimates of β for the $k-1$ temperature groups.

Each point shows how much larger the departure-probability gap between prisons without and with air conditioning is in the labeled temperature bin than in the coldest bin ($< 60°$F). For example, the yellow point estimates show the percentage point gap in the probability of departing between prisons without and with air conditioning when average monthly temperatures are $\geq 80°$F compared to when temperatures are $< 60°$F. Positive values indicate that departures are more common in prisons without air conditioning under hotter conditions.[15]

We estimate our main model on four different subsets of the data: all employees, all frontline correctional officers, frontline correctional officers with five or fewer years of experience, and frontline correctional officers with six or more years of experience. The estimates using all employees and all frontline correctional officers are signed in the expected direction and near statistical significance. However, the coefficient magnitudes are largely similar across temperature bins. In other words, there is no evidence of a diminishing effect of air conditioning as temperatures cool among all employees or all frontline correctional officers. However, subsetting by years of experience, the results are in line with our expectations. Among the least-experienced frontline employees, the difference in the probability of departing between prisons without and with air conditioning is 0.33 percentage points larger in the hottest months than in the coolest. This equates to an approximately 12% increase in the probability of departing relative to the baseline mean departure probability ($\mu = 2.7$). The results dissipate, as expected, as temperatures cool. There is no statistically significant departure-probability gap between prisons without and with air conditioning when average monthly temperatures are 60-69°F versus $< 60°$F. We also show, as expected, that there is no temperature effect for the more experienced employees with over five years of experience, likely due to affected employees having already departed the workforce.

15. Full tabular results are shown in Table C.4.

Figure 4 – Among the Least Experienced, Departing in the Summer More Likely in Prisons without Air Conditioning. Each point shows how much larger the departure-probability gap between prisons without and with air conditioning is in the labeled temperature bin than in the coldest bin (< 60°F). 95% confidence intervals plotted alongside point estimates. The values of μ show the baseline probability of departing on the percentage point scale for the respective subset of employees. Full tabular results are shown in Table C.4.



These results are largely robust to different sample inclusion criteria and definitions of experience. In Table C.1 we re-estimate Formula 1 with different definitions of which prisons should be counted as (not) having air conditioning. In addition to the results using our main operationalization, we show our main effect estimates when a prison is treated as (not) having air conditioning if its share of beds with air conditioning is (less) greater than 0.5 and when a prison is either fully air conditioned or lacks any beds with air conditioning. These two approaches allow us to, respectively, use our entire dataset and offer a sharper test of our main expectations. Our main findings in Figure 4 largely hold, with some caveats. Our main finding of a greater probability of departing among the least experienced employees in prisons without air conditioning in the hottest months of the year holds when the data is restricted to only prisons with full or no air conditioning. While still positively signed and similar in magnitude, the estimate does not retain statistical significance when using data from all prisons.

Our results also hold using cooling degree days in place of our binned temperature measure. For a given day, if the average temperature is above 65°F, the number of cooling degree days is equal to the difference between the average temperature and 65°F. If the average temperature is less than 65°F, the number of cooling degree days is equal to zero. Summed over the course of a month within a county, cooling degree days helps pick up on differences across months with the same mean temperatures but different variances. Helpfully, for our case, cooling degree days is also frequently used as a measure of demand for air conditioning (National Weather Service 2023). In Figure C.2 in the SI, we show the results of running Equation 1 with *temp* replaced by the number of cooling degree days for the county and month. The results, in line with those in Figure 4, suggest that the departure-probability gap between prisons without and with air conditioning widens as the number of cooling degree days increases.

Finally, our results are not an artifact of how we subset the data by years of experience. In Figure 4 we split the dataset into two levels by experience: frontline correctional officers with five or fewer years of experience and employees with six or more years of experience. In Figure C.1 we show that our estimate of a wider departure-probability gap across prisons

without and with air conditioning in hotter versus cooler months largely holds across different subsets of less-experienced employees.

6 The Cost of Installing Air Conditioning

The results above suggest that installing air conditioning in TDCJ prisons would improve the retention of correctional officers. Among other potential benefits of installing air conditioning (e.g., improved safety of incarcerated people and reduced court costs from heat-related litigation), the TDCJ and state stand to gain from the reductions in employee turnover facilitated by installing air conditioning. Current high levels of employee turnover in the TDCJ interfere with productivity and safety, require increased spending on recruiting and training new hires,[16] and necessitate expensive mandatory overtime.[17] The TDCJ is aware of these costs and the potential benefits of installing air conditioning. The former Executive Director of the TDCJ testified in court that he "absolutely" wants the entire TDCJ system to be air conditioned (Pitman 2025, p. 86).

Yet, installing air conditioning is expensive. As of August 2024, The TDCJ has approximately 95,000 uncooled beds in facilities with partial or no air conditioning. Many of these beds are located in prisons built well before the invention of modern heat mitigation technologies, complicating the construction process. In this section, we conduct back-of-envelope cost estimates of installing air conditioning throughout the TDCJ system. In order to provide context for these cost estimates, we also estimate the cost of an alternative, frequently used policy option for improving employee retention: raising salaries. We demonstrate that, relative to a salary increase that reduces turnover by the same amount as installing air conditioning, installing air conditioning is a relatively cost-effective policy for addressing understaffing.

There is no consensus about how much it would cost to install air conditioning throughout all TDCJ facilities. As a result, we use estimates from the TDCJ, legislative proposals, and

16. The state estimates that advertising for a correctional officer position and training a new hire costs approximately $10,000 (Sunset Advisory Commission 2025).

17. The TDCJ spent $277 million on overtime in fiscal year 2023 (Sunset Advisory Commission 2025).

prior air-conditioning projects to generate four different cost scenarios. The first scenario is from a 2023 TDCJ estimate of the cost of implementing a legislative bill mandating air-conditioning in all TDCJ facilities. The agency estimated that installation would cost over $1.1 billion, with annual maintenance costs totaling $20 million (McGinty 2023). This estimate is largely in line with an earlier estimate from 2019 that also placed the cost of installation at over $1 billion (McCullough 2019a). Critics of the 2019 estimates suggested that the agency had a history of over-estimating the costs of air conditioning. For instance, during a lawsuit over excessive heat in the Pack Unit the TDCJ initially stated that adding air conditioning to the facility would cost $20 million, but costs ended up totaling $4 million.

The second and third scenarios use appropriations and actual cost data to estimate installation costs per bed. The TDCJ was recently appropriated $118 million to install air conditioning covering 18,000 beds, at a cost of $6,556 per bed. Applying the $6,556/bed cost to the 95,000 beds that remain uncooled, installing air conditioning in this scenario would cost approximately $623 million. The third scenario uses actual cost data from three construction projects. In 2018, the TDCJ added air conditioning to the Pack and Hodge units due to a court order, at a cost of, respectively, $4 million and $2 million (McCullough 2018b). Likewise, in fiscal year 2021, the TDCJ air-conditioned 3,407 beds at a cost of $13.4 million. Applying the cost per bed of these three projects (in 2025 dollars) generates a cost estimate of $383 million. Finally, the fourth scenario uses amounts from a 2025 Texas House bill that called for the state to spend $300 million to install air conditioning in all TDCJ facilities over the next seven years (Texas Legislature 2025).

Table 1 – Cost Scenarios. Shows the upfront installation and annual maintenance costs of installing air conditioning under four scenarios. All dollar amounts are in millions of 2025 dollars. Under scenarios 2-4, annual maintenance costs are estimated as 1.6% of the initial installation costs. This is the ratio of annual to initial costs in a recent report outlining the costs of installing air conditioning in prisons operated by the Florida Department of Corrections.

Scenario	Description	Initial Cost (C_{cap})	Annual Cost (C_{oper})
1	2023 TDCJ Estimate	$1,182	$20
2	2025 Appropriation	$623	$10
3	Installations	$383	$6
4	H.B. 3006	$300	$5

Table 1 overviews the cost of installing air conditioning in the four scenarios. Outside of the first funding scenario where the TDCJ provided an estimate of the cost of annual maintenance, we assign the annual operating cost C_{oper} to be 1.6% of the initial capital cost of installation, which is the ratio of annual maintenance costs to upfront air-conditioning installation costs in a recent report prepared for the Florida Department of Corrections (KPMG 2023). We annualize these initial and annual costs using Formula 3. Solving the equation requires selecting a discount rate and expected lifespan of the air conditioning system. Rather than selecting a single value for each of these parameters, we estimate C_{annual} with inputs $\mathcal{R} \times \mathcal{L}$, where:

$$\begin{aligned} \mathcal{L} &= \{15, 20, 25, 30, 35, 40\}, \quad L \in \mathcal{L} \\ \mathcal{R} &= \{0.03, 0.04, 0.05, 0.06, 0.07\}, \quad r \in \mathcal{R} \end{aligned} \tag{2}$$

$$C_{\text{annual}} = \underbrace{C_{\text{cap}} \left(\frac{r(1+r)^L}{(1+r)^L - 1} \right)}_{\text{Annualized capital cost}} + \underbrace{C_{\text{oper}}}_{\text{Annual operating cost}} \tag{3}$$

$$\text{where } r = \text{discount rate}, \quad L = \text{system lifespan (years)}$$

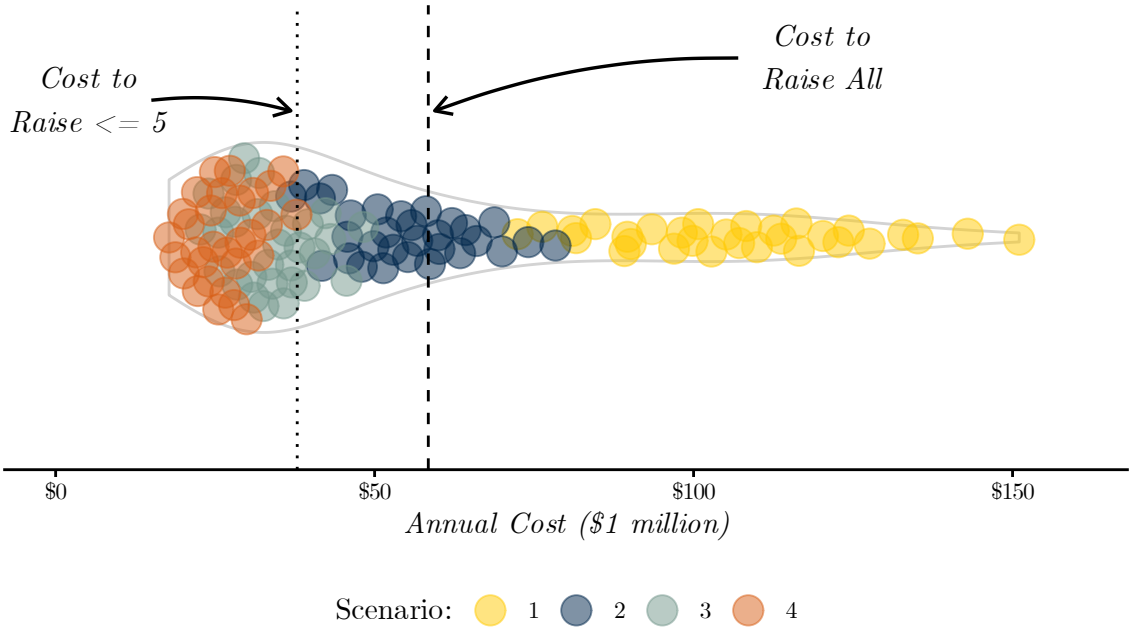
In order to benchmark our cost estimates, we estimate the cost of raising employee salaries by the amount necessary to reduce turnover by the same amount as installing air conditioning. We estimate the cost of two types of salary raises — one that applies to all frontline correctional officers in prisons lacking full air conditioning coverage and the other to frontline correctional officers with five or fewer years of experience in prisons lacking full air conditioning coverage. Salary raises are a favored policy tool to improve employee retention in the TDCJ. The agency has described the 15% pay raise given to correctional officers in August 2022 (shown in Figure 3) as having had "the most significant impact on Correctional Officer staffing levels compared to any other hiring initiative" (Collier 2024b). Since 2022, correctional staff in the TDCJ have received four raises, increasing salaries by 40% over the period.

We estimate the cost of raising employee salaries by the amount necessary to generate the same reduction in turnover among frontline correctional officers as installing air conditioning. Averaging across the estimates of the seasonal effect of air conditioning shown in Figure 4 and Table C.1, we calculate the mean difference in seasonal turnover across prisons with and without air conditioning weighted by the share of months in our dataset that fall within each temperature bin.[18] We calculate these values separately for all frontline correctional officers and frontline correctional officers with five or fewer years of experience. Using these estimates and the respective average marginal effects of increasing monthly salaries on the probability of departing (estimated via Formula 1), we find that raising salaries by 6.0% and 7.4% correlate with the same reduction in turnover as installing air conditioning for, respectively, all and the least-experienced frontline correctional officers.[19]

18. We assume zero difference in turnover between prisons with and without air conditioning during the coldest months of the year when average temperatures are less than 60°F.

19. See Section D in the Supplementary Information for more details on these calculations.

Figure 5 – Annualized Cost of Air Conditioning Similar to Equivalent Salary Raises. Shows the distribution of the estimated cost of installing air conditioning in all TDCJ facilities, by upfront cost scenario and different assumptions about system lifespan and the discount rate. The dashed line denotes the estimated cost of raising salaries of frontline correctional officers in prisons with partial or no air conditioning by 6.0%. The dotted line shows the estimated cost of raising salaries of frontline correctional officers with five or fewer years of experience by 7.4%. All amounts are in 2025 dollars.



The dashed line in Figure 5 shows the cost of raising salaries of all frontline correctional officers in prisons without full air conditioning by 6.0%. The dotted line shows the estimated cost of raising salaries for only those employees with five or fewer years of experience working in prisons without full air conditioning. The colored dots display the distribution of the annualized cost of installing air conditioning. The multiple dots within each scenario reflect the different assumptions regarding discount rates and the lifespan of the air conditioning system. Overall, the annual cost of installing air conditioning varies widely both within and across scenarios, ranging from around $18 million to $150 million annually (median = $42 million, mean = $55 million). In comparison, raising salaries for all and the least-experienced frontline correctional officers in prisons lacking full air conditioning costs, respectively, $58 million and $38 million per year. While expensive, installing air conditioning does not im-

pose prohibitive costs on the state. Its costs are in line with the costs of other policy tools frequently utilized to address understaffing.

Of course, caveats exist with these back-of-envelope cost estimates. For one, the benefits of air conditioning are not immediate. Unlike salary raises, which can be quickly implemented, approving the installation of air conditioning is not the same as actually cooling prisons. The construction process means that the benefits of air conditioning on turnover take time to accrue, which we do not account for in our estimates. In addition, we also treat the effect of air conditioning as constant across prisons that lack full air conditioning. This might not be the case, as the marginal benefits on turnover of, for example, installing 100 cooled beds might be less in a prison that is already nearly fully air-conditioned. While we do not account for this here, Figure 1 shows that most prisons without full air conditioning have minimal coverage, likely dulling the implications of this concern for our cost estimates.

7 Discussion

Reformers have long framed the lack of air conditioning in many U.S. prisons as a moral and constitutional issue. We offer a different, albeit complementary, perspective, arguing that poor working conditions are also a workforce issue. The street-level bureaucrats who work in prisons, like the incarcerated people they supervise, are negatively affected by excessive heat. However, unlike incarcerated people, correctional officers have the option to leave, further exacerbating the dangerously high levels of turnover and understaffing that plague the American carceral state.

Using personnel and air conditioning data from the largest state prison system in the U.S., we show that excessive heat caused by a lack of air conditioning contributes to correctional officer turnover. Before the onset of COVID-19, the least experienced correctional officers in the TDCJ were more likely to leave in the hottest months of the year if they worked in a prison lacking air conditioning. In particular, we show that the departure-probability gap between prisons without and with air conditioning widens by 0.33 percentage points in the hottest months relative to the coldest months. This is a meaningfully large effect size,

equating approximately 12% of less-experienced employees' baseline probability of departing.

While we add to a growing literature on how excessive heat fosters dangerous conditions within U.S. prisons, we innovate by both focusing on correctional officers and examining the cost-effectiveness of investing in air conditioning. We demonstrate that, when it comes to reducing correctional officer turnover, installing air conditioning imposes slightly more or similar costs than raising employee salaries. The initial investments of adding air conditioning are high, but given the size of the correctional officer workforce in Texas and many other states, raising employee salaries also comes at a considerable cost. Furthermore, adding air conditioning has other financial benefits not taken into account in our model. For instance, with air conditioning installed, states no longer need to pay to defend themselves in court from heat-related lawsuits brought by incarcerated people.

Our work has a few key limitations, which should be improved upon in future work. First, additional work should analyze the effects of heat on turnover during and after the COVID-19 pandemic. We find that our results are largely isolated to before the onset of the COVID-19 pandemic. There are a few possible reasons for this that deserve additional study in future work. The pandemic brought unprecedented challenges to prisons, many of which were particularly affected by the virus. As a result, new unobserved confounders that are difficult to control for may be biasing our results and masking an underlying relationship between air conditioning and departing. The lack of considerable data from "after" the pandemic makes it difficult, in this study, to fully disentangle the possible reasons for the variation in our findings over time.

Future work should also investigate other possible responses to excessive heat among correctional officers and other types of street-level bureaucrats. Exiting the workforce is one, particularly extreme, measure that correctional workers can take in the wake of poor conditions. They can also take vacation days or be forced to work additional overtime shifts to cover vacancies. Future research should incorporate these other dependent variables. In addition, while Texas is a particularly salient case for studying the relationship between heat and correctional officer turnover, correctional officers in other states might be even more

responsive to heat. Climate change is increasing the number of hot summer days across the country in places where people are less used to excessive heat (Tuholske et al. 2024). As a result, correctional officers outside of the South are beginning to face increasing and unprecedented heat-related difficulties.

Overall, our findings underscore the need for additional investigations into how street-level bureaucrats' physical working environments affect their willingness to remain in their jobs. While the case of excessive heat in Texas' prisons is a particularly extreme case of poor working conditions, influential frontline public employees across the country work in substandard settings due to underinvestments, excessive heat, and other factors. By demonstrating the relationship between poor conditions and public sector turnover, our study highlights the importance of taking seriously the on-the-job safety and comfort of frontline public employees.

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Supplementary Information

Excessive Heat and Correctional Officer Turnover: Evidence from Texas

A Defining Employee Departures **SI-1**

B Heat in Texas **SI-2**

C Additional Results and Robustness Checks **SI-4**

D Cost Analysis **SI-11**

D.1 Estimating the Commensurate Percent Salary Raise SI-11

D.2 Estimating the Cost of the Percent Salary Raise SI-12

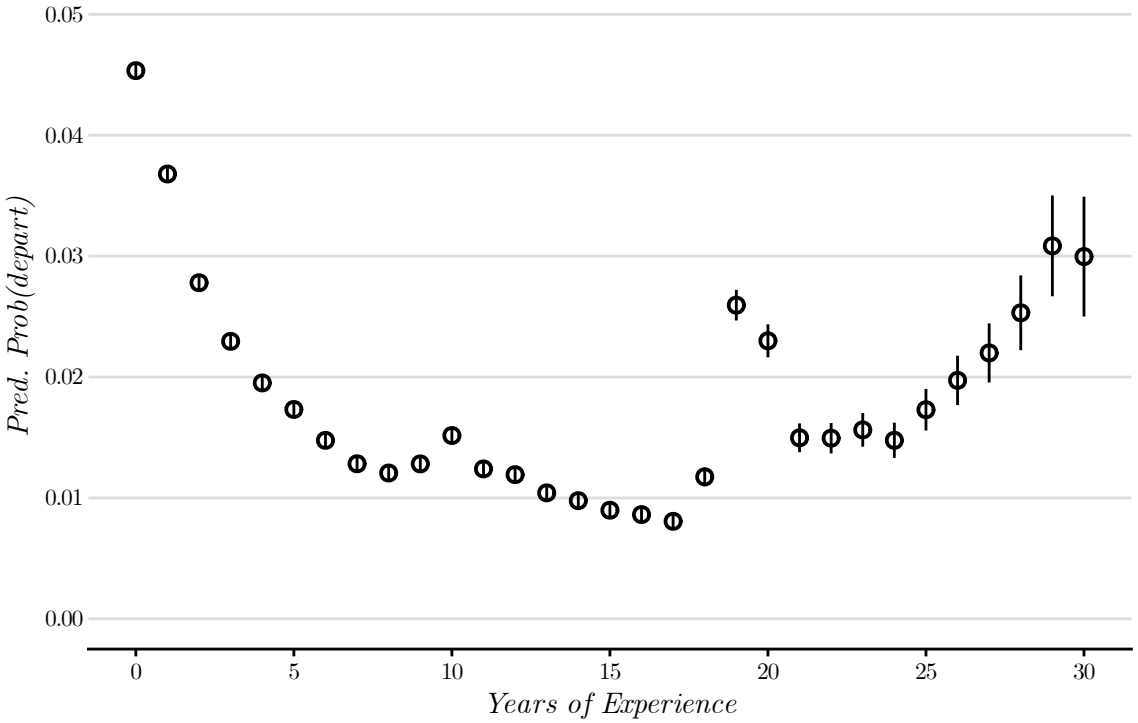
A Defining Employee Departures

The dependent variable in all analyses is whether an employee departs the TDCJ in a given month. We use information from two different sources to define an employee's month of departure. First, via the personnel dataset, it is straightforward to find the last month of observed continuous employment (i.e., the end of an employee's spell of employment with the TDCJ). However, since the personnel data is a snapshot of all TDCJ security employees on the last day of the given month, it is unclear whether someone actually departed on the last day of the month or at some point before the last day of the next month. The second source of information is an additional dataset provided to us by the TDCJ that lists all employees who departed during our study period and their actual separation date (which we truncate to the actual month of separation). While more precise, these data do not include instances of employees moving from security to non-security (e.g., case manager) roles within the TDCJ. Despite their differences, the two data sources are largely similar. Employees' last observed month of employment and actual month of separation have a Jaccard index value of 0.90.

Given the benefits and drawbacks of both data sources, we use a two-pronged approach to define employees' month of departure. First, via the additional data provided to us by the TDCJ, we merged in the actual separation date for all employees who departed during our study period. Second, if an employee spell ends before the last month that we have data (December 2022) without a separation, we deem it to be an instance of an employee moving to a non-security role within the TDCJ and assign it as a departure.

Figure A.1 shows the predicted probability of departing by years of experience, marginalizing over season and prison air conditioning status. There are two jumps in the probability of departing at around ten and twenty years of experience with the TDCJ. These are likely both the result of retirement benefit eligibility rules. Employees with ten years of experience can retire at age 65 with annuity and insurance benefits, while correctional officers and other public safety employees in Texas are eligible for expanded benefits with twenty years of service through the Law Enforcement Custodial Officers System. Some employees in our dataset are eligible for expanded retirement benefits after five years of service with the TDCJ, but there is no evidence of an increase in the probability of departing at that level of experience (Texas Department of Criminal Justice, Human Resources Division 2021; Employees Retirement System of Texas 2023).

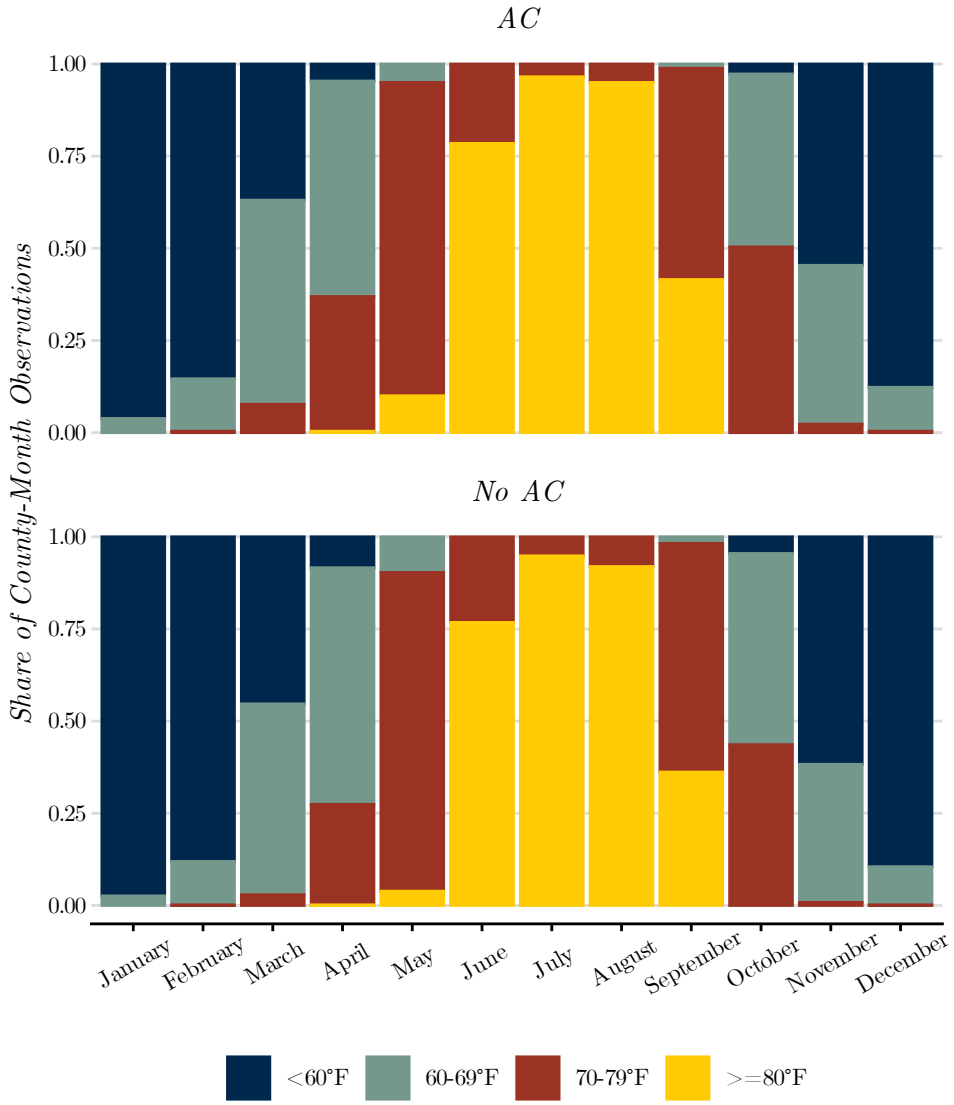
Figure A.1 – Overall Probability of Departing Increases at 10 and 20 Years of Experience Due to Vesting. Shows the predicted probability of departing by years of experience, averaged across seasons and prisons' air conditioning status. Predicted probabilities estimated using a logit model that regresses whether an employee departed on a years of experience fixed effect. The increases at ten and twenty years of experience are due to eligibility for expanded retirement benefits.



B Heat in Texas

Figure B.1 shows the monthly distribution of the binned temperature variable. Most of the highest-temperature months occur in June, July, and August. The coolest months largely occur in November, December, January, and February.

Figure B.1 – Distribution of Temperature by Month of Year. Shows the share of temperature readings by month of the year. The data includes all county-month observations from 2010-2023 for TDCJ facilities that have either $> 90\%$ or $< 10\%$ air conditioning coverage.



C Additional Results and Robustness Checks

Figure C.1 – Main Findings Largely Robust to Different Experience Subsets. Re-estimates the results in Figure 4 using different subsets of the data. Going from the top to the bottom, the results show how much larger the departure-probability gap between prisons without and with air conditioning is in the labeled temperature bin than in the coldest bin (< 60°F) among frontline correctional officers with 0-1, 0-3, 0-5, and 0-7 years of experience. The bottom results show the estimates for all frontline correctional officers.

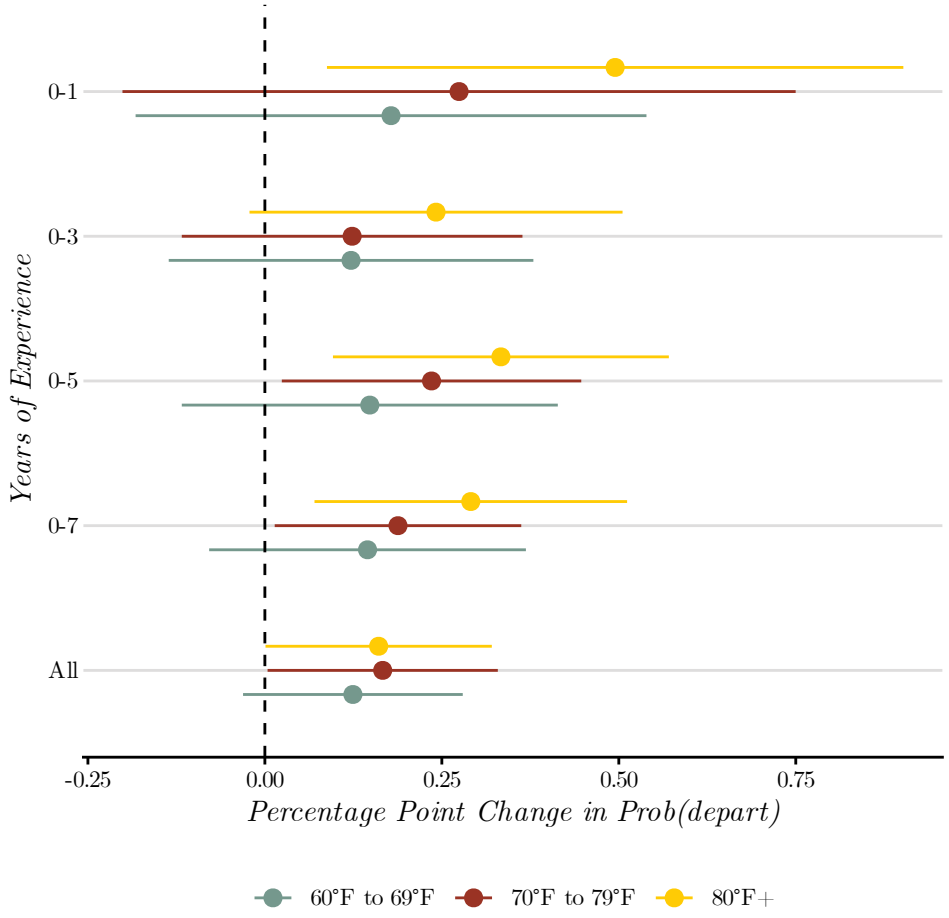


Table C.1 – Effect Estimates of Interest (Before March 2020). Shows the results of our main model run separately on different groups of employees and different definitions of which prisons are deemed as having air conditioning. Controls and fixed effects are omitted and standard errors are clustered by prison.

	All Employees	All Frontline	Frontline (0-5 years)	Frontline (6+ years)
Main Results (prison has (no) AC if share of beds >0.9 (<0.1)				
80°F+ × No AC	0.13	0.16	0.33**	-0.06
	(0.08)	(0.08)	(0.12)	(0.09)
70-79°F × No AC	0.16	0.17*	0.24*	-0.00
	(0.08)	(0.08)	(0.11)	(0.09)
60-69°F × No AC	0.10	0.12	0.15	0.01
	(0.08)	(0.08)	(0.14)	(0.08)
All Prisons (prison has (no) AC if share of beds >0.5 (<0.5)				
80°F+ × No AC	0.18*	0.19*	0.24	0.02
	(0.08)	(0.08)	(0.15)	(0.08)
70-79°F × No AC	0.20**	0.20**	0.26*	0.02
	(0.07)	(0.07)	(0.10)	(0.08)
60-69°F × No AC	0.14*	0.15*	0.19	0.06
	(0.06)	(0.06)	(0.11)	(0.07)
Sharpest Comparison (prison has (no) AC if share of beds == 1 (==0)				
80°F+ × No AC	0.26***	0.31***	0.49***	0.00
	(0.06)	(0.07)	(0.14)	(0.09)
70-79°F × No AC	0.30***	0.32***	0.41**	0.07
	(0.07)	(0.07)	(0.12)	(0.09)
60-69°F × No AC	0.19*	0.21*	0.29	-0.03
	(0.08)	(0.09)	(0.15)	(0.09)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.2 – Effect Estimates of Interest (All Data). Shows the results of our main model run separately on different groups of employees and different definitions of which prisons are deemed as having air conditioning. Controls and fixed effects are omitted and standard errors are clustered by prison.

	All Employees	All Frontline	Frontline (0-5 years)	Frontline (6+ years)
Main Results (prison has (no) AC if share of beds >0.9 (<0.1)				
80°F+ × No AC	0.08	0.11	0.32*	-0.14
	(0.06)	(0.06)	(0.13)	(0.08)
70-79°F × No AC	0.21*	0.22*	0.33**	-0.00
	(0.08)	(0.09)	(0.11)	(0.10)
60-69°F × No AC	0.07	0.11	0.14	-0.00
	(0.07)	(0.08)	(0.14)	(0.08)
All Prisons (prison has (no) AC if share of beds >0.5 (<0.5)				
80°F+ × No AC	0.15*	0.16*	0.27	-0.06
	(0.06)	(0.07)	(0.15)	(0.07)
70-79°F × No AC	0.27***	0.27***	0.38***	0.04
	(0.07)	(0.08)	(0.10)	(0.08)
60-69°F × No AC	0.11	0.14	0.15	0.05
	(0.06)	(0.07)	(0.10)	(0.07)
Sharpest Comparison (prison has (no) AC if share of beds == 1 (==0)				
80°F+ × No AC	0.15*	0.19*	0.36*	-0.12
	(0.06)	(0.07)	(0.16)	(0.09)
70-79°F × No AC	0.32***	0.34***	0.49***	0.03
	(0.07)	(0.08)	(0.11)	(0.10)
60-69°F × No AC	0.12	0.16	0.23	-0.06
	(0.09)	(0.09)	(0.16)	(0.09)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.3 – Effect Estimates of Interest (After March 2020). Shows the results of our main model run separately on different groups of employees and different definitions of which prisons are deemed as having air conditioning. Controls and fixed effects are omitted and standard errors are clustered by prison.

	All Employees	All Frontline	Frontline (0-5 years)	Frontline (6+ years)
Main Results (prison has (no) AC if share of beds >0.9 (<0.1)				
80°F+ × No AC	-0.08	-0.03	0.40	-0.42*
	(0.22)	(0.25)	(0.55)	(0.19)
70-79°F × No AC	0.38**	0.39*	0.76*	0.04
	(0.14)	(0.17)	(0.29)	(0.21)
60-69°F × No AC	0.00	0.10	0.28	-0.02
	(0.21)	(0.22)	(0.47)	(0.16)
All Prisons (prison has (no) AC if share of beds >0.5 (<0.5)				
80°F+ × No AC	-0.02	0.01	0.33	-0.40*
	(0.17)	(0.19)	(0.36)	(0.17)
70-79°F × No AC	0.48**	0.46**	0.74*	0.12
	(0.14)	(0.16)	(0.32)	(0.17)
60-69°F × No AC	-0.05	0.01	0.02	-0.02
	(0.19)	(0.20)	(0.37)	(0.14)
Sharpest Comparison (prison has (no) AC if share of beds == 1 (==0)				
80°F+ × No AC	-0.24	-0.25	-0.09	-0.48
	(0.25)	(0.28)	(0.53)	(0.25)
70-79°F × No AC	0.35*	0.35	0.71*	-0.05
	(0.17)	(0.21)	(0.35)	(0.25)
60-69°F × No AC	-0.09	0.05	0.20	-0.10
	(0.22)	(0.24)	(0.46)	(0.21)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.4 – Full Tabular Results. Shows the estimates for our main model, estimated using all employees, all frontline correctional officers, frontline correctional officers with 5 or fewer years of experience, and frontline correctional officers with 6 or more years of experience. Data is restricted to before the onset of the COVID-19 pandemic. A prison is defined as (not) having air conditioning if the share of beds covered by air conditioning exceeds 0.9 (is less than 0.1)

	All Employees	All Frontline	Frontline (0-5 years)	Frontline (6+ years)
80°F+ × No AC	0.13	0.16	0.33**	−0.06
	(0.08)	(0.08)	(0.12)	(0.09)
70-79°F × No AC	0.16	0.17*	0.24*	−0.00
	(0.08)	(0.08)	(0.11)	(0.09)
60-69°F × No AC	0.10	0.12	0.15	0.01
	(0.08)	(0.08)	(0.14)	(0.08)
80°F+	−0.02	−0.07	−0.19	0.00
	(0.13)	(0.15)	(0.22)	(0.16)
70-79°F	−0.15	−0.16	−0.22	−0.07
	(0.13)	(0.13)	(0.18)	(0.16)
60-69°F	−0.07	−0.09	−0.05	−0.11
	(0.09)	(0.09)	(0.15)	(0.10)
80°F+ × Urban	−0.31***	−0.33***	−0.40***	−0.22*
	(0.07)	(0.08)	(0.10)	(0.08)
70-79°F × Urban	−0.06	−0.08	−0.03	−0.06
	(0.06)	(0.06)	(0.09)	(0.07)
60-69°F × Urban	−0.05	−0.07	−0.04	−0.04
	(0.06)	(0.07)	(0.11)	(0.07)
N Ag. and Constr. Jobs × No AC	−0.00	−0.00	−0.00	−0.00
	(0.00)	(0.00)	(0.00)	(0.00)
N Ag. and Constr. Jobs	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)
White	0.50***	0.55***	0.74***	0.28***
	(0.06)	(0.07)	(0.11)	(0.03)
Male	−0.27***	−0.26***	−0.35***	−0.18***
	(0.03)	(0.04)	(0.06)	(0.03)
Age	−0.03***	−0.03***	−0.06***	0.02***
	(0.00)	(0.00)	(0.00)	(0.00)
log(Monthly Salary)	−2.29***	−2.11***	−2.53***	−0.91***
	(0.17)	(0.21)	(0.33)	(0.15)
Unemployment Rate	−0.08***	−0.10***	−0.12***	−0.08**
	(0.02)	(0.02)	(0.03)	(0.02)
log(Incarcerated Pop.)	0.01	−0.00	0.09	−0.08
	(0.07)	(0.07)	(0.09)	(0.09)
log(Oil Production)	0.03	0.03	0.03	0.00
	(0.04)	(0.04)	(0.05)	(0.03)
log(Gas Production)	−0.06*	−0.05*	−0.07*	0.00
	(0.03)	(0.03)	(0.04)	(0.03)
Facility FEs	✓	✓	✓	✓
Month FEs	✓	✓	✓	✓
Years of Experience FEs	✓	✓	✓	✓
Num. obs.	1872559	1659749	896515	763234

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Figure C.2 – Main Findings Robust to Cooling Degree Days Rather than Binned Temperature. Re-estimates the results in Figure 4 using cooling degree days in place of binned temperature. Cooling degree days equal the sum, over each day in the month, of the amount by which the county's average daily temperature exceeded 65°F (days at or below 65°F count as zero). In line with the results in Figure 4, we find that the departure-probability gap between prisons without and with air conditioning widens as the number of cooling degree days increases.

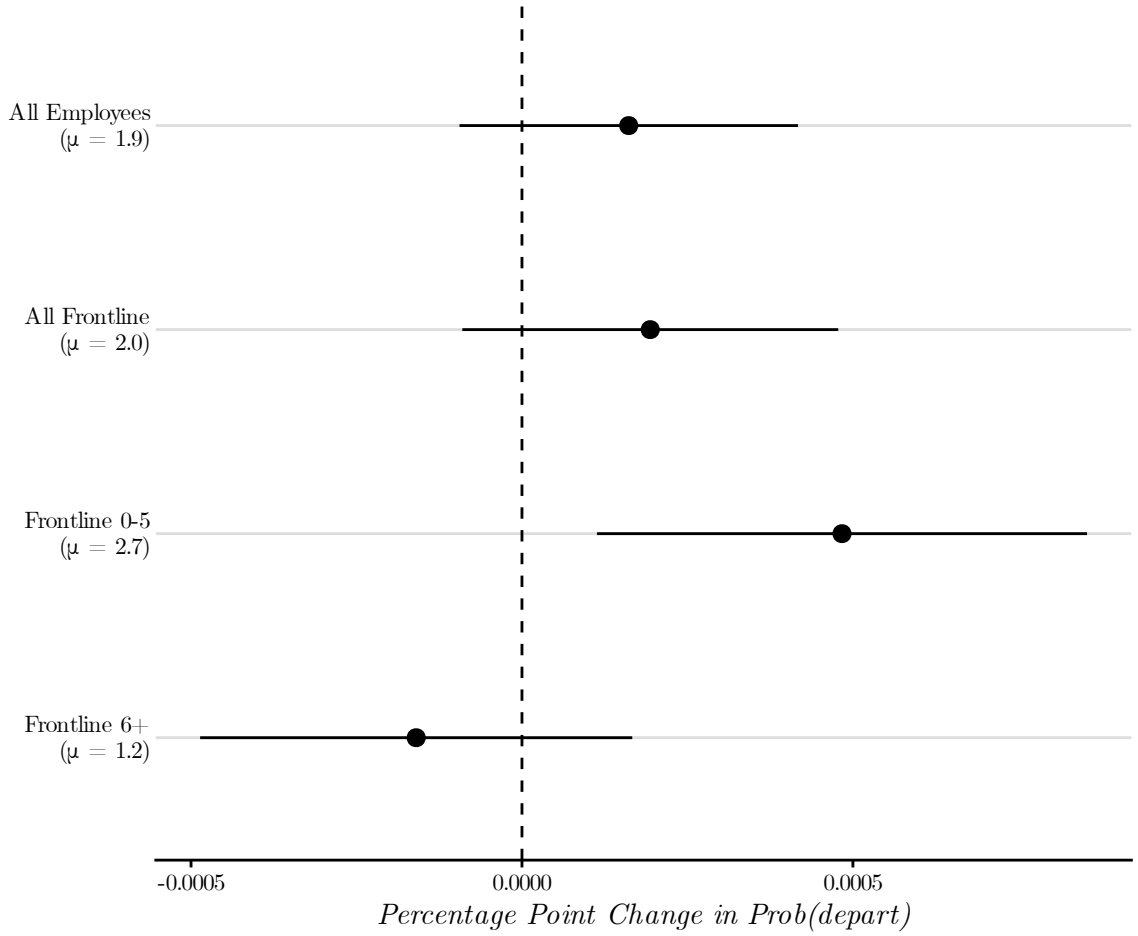


Figure C.3 – Switching Prisons is Correlated with Years of Experience, but Little Evidence that Switching From a Prison Without Air Conditioning to One With Air Conditioning is Correlated with Experience. Shows the predicted probability of switching from one prison to another (yellow), from a prison without air conditioning to one with air conditioning (maroon), and from a prison with air conditioning to one without air conditioning (blue). Results are from a logistic regression of the respective switching indicator against binned years of experience. The predicted probabilities are calculated while clustering standard errors at the employee level. Overall, the results show that the predicted probability of switching from any prison to another regardless of air conditioning decreases with years of experience, before increasing again for the most experienced employees. There is no evidence that similar results hold for switching out of prisons with air conditioning and the results are much more muted for switching from a prison without air conditioning to a prison with air conditioning.



D Cost Analysis

In this section we describe how we estimate the cost of raising employee salaries by the amount necessary to reduce turnover by the same amount as installing air conditioning. We estimate the cost of raising salaries for all frontline correctional officers employed in prisons with partial or no air conditioning by the percent necessary to reduce turnover by the same amount as installing air conditioning.

D.1 Estimating the Commensurate Percent Salary Raise

We first estimate the percent change in salary necessary to reduce the probability of departing in a given month by the same amount as installing air conditioning.

The annual predicted probability of a frontline correctional officer departing at baseline and with air conditioning installed is defined as a function of the baseline monthly probability of departing (p_m) and the difference in the probability of departing under air conditioning (Δ_{ac}):

$$p_a = 1 - (1 - p_m)^{12}$$
$$p_a^{ac} = 1 - (1 - (p_m + \Delta_{ac}))^{12}$$

We are interested in solving the following identity, where p_a^x is the annual predicted probability of departing when monthly salaries increase by x percent.

$$p_a^x - p_a = p_a^{ac} - p_a$$

Where:

$$p_a^x = 1 - (1 - (p_m + x \times \Delta_{salary}))^{12}$$

Rearranging terms:

$$0 = (1 - (1 - (p_m + \Delta_{ac}))^{12}) - (1 - (1 - (p_m + x \times \Delta_{salary}))^{12})$$

We solve for x using three inputs, which are shown below in Table D.1. We estimate the change in the probability of departing when air conditioning is present by finding the mean of the effect of departing in a prison without air conditioning by temperature bin across the different models in Figure 4 and Table C.1. We assume zero difference in departures across prisons with and without air conditioning in the coldest temperature bin and multiply the estimates by -1 to get the estimates of the effect of installing air conditioning. We then weight these mean estimates of the difference in departures across prisons with and without

air conditioning by temperature bin by the number of months in the dataset that fall within each of these temperature bins. The estimate of the change in the probability of departing when monthly salaries increase by one unit is calculated as the correlation between departing and logged monthly salaries, averaged across the models in Figure 4 and Table C.1. p_m is the mean probability of departing in a given month for all frontline correctional officers before the onset of COVID-19.

Table D.1 – Inputs and Output of Percent Change in Salary. Shows the inputs used to define the percent change in monthly salary necessary to generate the same reduction the probability of departing as installing air conditioning. The percent change estimate is shown in the final column. Note that we multiply the values for Δ_{ac} by -1 because the models estimate the effect of no air conditioning. We divide the values for Δ_{salary} by 100 because they are on the log scale. Dividing by 100 gets the Δ_{salary} estimate to show the effect of increasing salary by 1%.

	Inputs			
	p_m	Δ_{salary}	Δ_{ac}	x
All frontline COs	0.023	-0.023/ 100	-1×0.0014	6.0
Frontline COs with $\leq$5 Years Exp.	0.031	-0.027/ 100	-1×0.0020	7.4

D.2 Estimating the Cost of the Percent Salary Raise

Now that we know the percent change in monthly salary necessary to reduce turnover by the same amount as installing air conditioning, we estimate the annual cost of the raises in inflation-adjusted 2025 dollars:

$$raise_a = cost_m \times scale \times 12 \times x \div 100$$

Where $cost_m$ is the total salary paid to frontline correctional officers in prisons with partial or no air conditioning in January 2023 (the last month we have data) and x is the percent raise necessary to match the effect of air conditioning on turnover. *scale* accounts for recent increases in salaries and staffing since January 2023. A recent report from the TDCJ notes that 18,757 frontline correctional officers work in the state (an increase of 9 percent from the total number in our data in January 2023). Likewise, according to a recent statement by the TDCJ, salaries have increased 40% since 2022, including a 15% raise in August 2022. Table D.2 lists the input values and corresponding estimates of the annual cost of the salary raises. As a spot check, we note that the TDCJ's budget request to the legislature for Fiscal Years 2026-2027 estimates that raising salaries for all correctional officers (including those employed in air-conditioned facilities) by 10% costs $165 million per year (Collier 2024a), which is largely in line, proportionally, with our estimate of the cost of a 6.0% raise for employees in facilities with no or partial air conditioning.

Table D.2 – Cost of Salary Raises. Shows the inputs used to estimate the cost of the salary raises necessary to generate the same reduction the probability of departing as installing air conditioning. All dollar amounts are in millions of 2025 dollars and are calculated only for frontline correctional officers working in prisons with partial or no air conditioning.

	Inputs			
	$cost_m$	$scale$	x	$raise_a$
All frontline COs	\$59.2	1.09×1.25	6.0	\$58.4
Frontline COs with $\leq$5 Years Exp.	\$31.4	1.09×1.25	7.4	\$37.9